This system describes the behavior of a particle with unit mass in one dimension, with energy function $E(x, \dot{x}) = \dot{x}^2/2 + \Omega^2 x^2/2$, where its motion is governed by a linear 2nd order equation $\ddot{x} + \Omega^2 x = 0$. The solution, for given $x(0) = \xi$, $\dot{x}(0) = v(0) = \eta$, is

$$x(t) = \xi \cos(\Omega t) + \frac{\eta}{\Omega} \sin(\Omega t).$$

This system, despite its simplicity, is an important test case for many schemes in molecular dynamics and will appear in many forms throughout this book (we generally refer to any system where the energy has quadratic position dependence as harmonic and any one degree of freedom harmonic system as a harmonic oscillator).

Example 1.4 (Single Degree of Freedom) Consider a simple system with a single degree of freedom (with unit mass $m = 1$ for simplicity) and energy function $E(x, \dot{x}) = \dot{x}^2/2 + U(x)$ (which includes the harmonic oscillator as a special case). The equations of motion are

$$\dot{x} = v, \qquad \dot{v} = -\frac{\partial U}{\partial x}.$$

Solutions are smooth functions $(x(t), v(t))$ that evolve on curves that are defined by $E(x, v) \equiv \mathrm{E}$, where E is the energy calculated at the initial point $(x(0), v(0)) = (\xi, \eta)$. Near this point, provided $\eta \neq 0$, let us solve the equation $E(x, v) = E(\xi, \eta)$ for v as a unique smooth function of x, ξ, η using the implicit function theorem: $v = V(x, \xi, \eta)$. Inserting this into the differential equations we then find

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v = V(x, \xi, \eta).$$

This is a separable differential equation; it is therefore easily integrated from any provided initial value, $x(0) = \xi$, as a function of t, resulting in a relation $x = X(t, \xi, \eta)$. Then $v = V(X(t, \xi, \eta), \xi, \eta)$ and we see that it is possible to derive the formula for the solution (x, v) as a function of t and the initial conditions. This representation provides a complete characterization of the *local* behavior of solutions, i.e., within a small interval of our starting point. To describe the *global* solutions, we would need to piece together several such local representations. For example if we used this approach to solve the harmonic oscillator, the equation for x would be $\dot{x} = \pm\sqrt{2\mathrm{E} - \Omega^2 x^2}$, and the sign that appears would define different parts of the solution which may be concatenated together to describe a global solution. "Degenerate" points, where $\partial E/\partial v = 0$, lead to difficulties that must be resolved on a case by case basis.

In more complicated systems, it is usually impossible to write down a formula for the solution in an explicit form; so approximation (typically using numerical methods) becomes essential to being able to predict dynamics.